DPUSGEVidGen

You can use DPUSGEVidGen to create digital video files during the simulation. The following elements can be inserted into the video:

- The contents of one SGEView **or** one BirdsEye **or** one MMI window
- Optionally, the image of an external camera can be inserted
- Additionally, a timecode can be inserted.

Most commonly, a SGEView is used as the source; this is assumed in the rest of the documentation for simplicity. Wherever the documentation mentions an SGEView, a BirdsEye or MMI window can be used instead.

The format of the recorded video file is Windows AVI (.avi).

- Any installed video codec can be used to compress the video file. Best results are usually achieved using current versions of the DivX codec or the x264vfw codec.
- The image size of the recorded video is automatically selected such that all information is visible.
- The video frame rate is based on the SGE view's update rate. It can be reduced by the proper divisor *Subsample* in order to reduce the CPU and hard disk load.

1. Dynamic parameters:

Warning: The parameters are only effective if an ExternalSource has been specified.

SGEOnTop	Should the SGE be on top (*true*) or below (*false* = default value) the external source?
SGEDownscale	Reduces the SGE image size by a proper divisor (1 = default = original size, 2 = half size, …)
SGEOpacity	Opacity of the SGE graphics (0 = not visible at all, 128 = semitransparent, 256 = completely opaque)
SGEX, *SGEY*	Position of the top-left edge of the SGE graphics within the created video
ExternalDownscale	Reduces the camera image size by a proper divisor (1 = default = original size, 2 = half size, …)
ExternalOpacity	Opacity of the camera image (0 = not visible at all, 128 = semitransparent, 256 = completely opaque)
ExternalX, *ExternalY*	Position of the top-left edge of the camera image within the created video

Timecode	Should a time code be inserted into the video (*true*, the default) or not (*false*).
TimecodeX, *TimecodeY*	Position of the top-left edge of the time code within the created video.

2. Static parameters:

InternalSource	Instance name of the DPU whose graphics output is recorded. It can be an instance of DPUSGEView, DPUBirdsEye or DPUMMI. The default value is "SGEView", which is the instance name of the SGE view that is defined in SILAB's default configuration.
Subsample	Proper divisor of the frame rate. The following equation is valid: Video frame rate = SGEView update rate / *Subsample*. Default: 1
VideoCodec	Name of the video codec that is used for compression. Default value: "*DivX*". If the name is empty (*VideoCodec* = ""), uncompressed video will be created (this is not recommended).
Quality	Quality setting for the video codec (range: 0..100, default: not set). Most video codecs ignore the quality setting.
Grayscale	The default value is 0, i.e. the video is saved in full colour. If set to 1..3, a greyscale output is created. Usually, video codecs do not accept greyscale input, so this will only work if no *VideoCodec* is specified. In detail, the following formats are used: Grayscale=1 Format YUY2 (16 bits/Pixel) Grayscale=2 Format Y800 (8 bits/Pixel) Grayscale=3 Format RGB8 (8 bits/Pixel). The format YUY2 saves 8 bits of greyscale information per pixel, but wastes another 8 bits per pixel. The formats Y800 and RGB8 are more efficient (i.e. no waste bits), but many video applications do not support them.
KeyFrameRate	Specifies the number of key frames in the created video. Values around 30 are recommended (i.e. about one key frame per second). If not specified, the video codec's default setting is used.
CompressionOptions	File which contains additional compression options for the video codec. If the file does not exist, a dialog is opened in which the options can be configured. They are then saved into this file. If a relative path is specified, the file is searched / created below SILAB's installation path.
Path	Output path of recorded file. If no absolute path is provided the path starts in the installation path of SILAB. Default: "*Data*\"
SaveFile	File name of the created video (within the Path folder). Default: "*SGEVidGen*" Alternatively, it can be specified using the launch data *VAR_DATAFILENAME*.
Suffix	Suffix that is inserted into the file name (after *SaveFile*, but before *Extension*). Default: (*none*)

Extension	File name extension (without leading dot). Default: "*avi*". It is only appended to the file name if *SaveFile* doesn't already contain an extension.
Overwrite	If set to *true*, an already existing video file is overwritten. If set to *false* (the default), an error message is issued and the simulation cannot be started if the output file already exists.
DiskFree_Warning_MB	Minimum recommended free disk space (MB) before starting a simulation. If less space is free, a warning is issued. Default: 10000
DiskFree_Error_MB	Minimum required free disk space (MB) before starting a simulation. If less space is free, the simulation cannot be started, because the space will probably run out. Default: 500
Preview	If set to *true*, a preview window is shown which displays the output video. If set to *false* (the default), the video is only written to the file. The preview window should only be used to control the video image. It is recommended to disable the preview during the actual recording because the preview creates additional CPU load.
PreviewX, *PreviewY*	Position of the preview window on screen (in pixels).
PreviewW, *PreviewH*	Size of the preview window on screen (in pixels).
ExternalSource	If set to true, additionally to the SGE view, the image from an external video source (e.g. a webcam or from a video capture card) is inserted into the output video. Default: *false*
VideoSource	Can be used to select an external video source, if more than one is connected to the PC. Specify the name of the desired video source (or an unambiguous part of it). The names of all connected video sources are displayed if *VideoSource=XXX* is specified.
AudioSource	Can be used to select an audio source, if more than one is connected to the PC. Specify the name of the desired audio source (or an unambiguous part of it). The names of all connected audio sources are displayed if *AudioSource=XXX* is specified.
ExternalW, *ExternalH*	Desired width and height of the video captured from the external camera. If not specified, a camera-dependent default value is used. **Warning**: Most cameras accept only a number of fixed video formats. A similar format will be selected if the desired format is not available. Additionally, some formats may not work correctly although they are advised by the camera's driver.
TimecodeBitmap	Image for the timecode generation (it is searched below SILAB's install path). A TGA or PNG image with alpha channel can be used. Default: *lib\graphics\default\images\timecode24.png*
SyncReactSpeed	Specifies how quickly the A/V synchronization reacts to changes in the SGE frame rate.

	Valid values are 0.1 up to 0.9. Default: 0.5, a good compromise between fast synchronization and changing audio pitch.
AudioOffsetMS	Can be used to correct a sync error in the created video file. This can especially happen if separate cards are used for audio and video capture (e.g. a discrete video capture card and the PC's built-in sound card). Positive values play audio earlier, negative values play it later.

3. Outputs:

VideoTimeMS	Current video time stamp, in milliseconds. If a DPUDataFile recording is running in parallel, it is recommended to add this variable to the DPUDataFile, so that the files can be synchronized precisely afterwards.
NDroppedFramesCurrent	Counts the number of video frames that had to be dropped within the last second due to CPU overload.
NDroppedFramesTotal	Counts the number of video frames that had to be dropped during the whole simulation due to CPU overload. This value should always be less than five; otherwise some parameters must be reduced.

4. Dependencies to other DPUs:

DPUSGEVidGen must run together with the graphics DPU whose image is recorded. The Index value of **DPUSGEVidGen** must be higher than the Index value of the graphics DPU.

5. Recommendation:

- Computing the simulation's 3D graphics and encoding the video at the same time imposes high demands on the computer. A quad-core processor is recommended.
- To minimize stuttering in the created video file, the frame rate of the SGE graphics and the external camera should be the same. For example, if the external camera delivers pictures with 25 Hz frame rate (PAL standard), the computer should be set to a *Frequency* of 50 Hz in the *Computerconfiguration*, and the *Subsample* parameter of DPUSGEVidGen should be set to 2. Thus, both frame rates are 25 Hz.

6. Example:

The DPU works well when using the following script:

```
DPUSGEVidGen SIGVidGen
{
	Computer = {RECORD};
	Index = 100;

	VideoCodec = "DivX 6";
	CompressionOptions =
		“lib\configurations\VidCap\DivX6HiRes.vcc";
```

```
        Overwrite = true;
        SaveFile = "DATA\SGEVidGen.avi";

        Subsample = 2;

        Preview = true;
        ExternalSource = true;

        PreviewX = 650;
        PreviewY = 0;

        ExternalDownscale = 2;
        ExternalOpacity = 220;
        ExternalX = 320;
        ExternalY = 240;
};

RECORD.SIGView.WindowW = 640;
RECORD.SIGView.WindowH = 480;
RECORD.SIGView.VerticalSync = false;
```